

Module 6: Lesson 3

Multi-Armed Bandits: Non-Stationary Case



Outline

- ▶ Non-Stationary k -Armed Bandit problem: Actions and learning.
- ▶ Results from a k -Armed Bandit testbed.

A non-stationary k -Armed Bandit problem: Actions and learning

In Lesson 3, we are going to implement the learning methodologies for the k -Armed Bandit in the non-stationary case.

Non-stationarity means that the actual average rewards can change over time. Non-stationarity is the case most commonly encountered in reinforcement learning in general, and finance in particular.

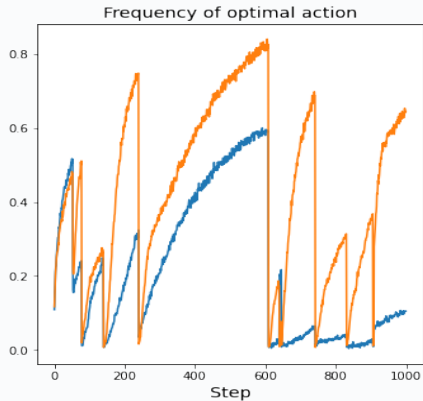
We aim at comparing the performance of the updating policy that averages across the past history of rewards and the updating policy that uses a fixed parameter $\alpha = 0.2$.

We are going to fix an ϵ -greedy policy that uses $\epsilon = 0.1$.

To make the problem non-stationary, we determine some random dates at which a "regime change" takes place, where the average reward is drawn again from a standard normal distribution.

We set in advance a one percent probability for the occurrence of a new regime.

A stationary k -Armed Bandit problem: Results



A stationary k -Armed Bandit problem: Results

In general, the policy that exploits a fixed parameter α to update the expected rewards behaves similar to a greedy action, which learns quick after changes in regime.

Thus, after regime changes, it tends to outperform the averaging method whose estimates are more dependent on the past.

The outperformance of one method over the other also depends on how the regime changes alter the ordering of actions in terms of expected rewards and the duration of regimes.

If the persistence in the ordering of actions is low, or even negative, the α -updating method may be superior.

Even in the case where there is a positive correlation in the ordering, the α -updating method may perform better because it usually leads to greater exploration.

Summary of Lesson 3

In Lesson 3 we have looked at:

- ▶ Understand the workings of a k -Bandit problem in a non-stationary setup.
- ▶ Understand the trade-offs involved in the choice of the α -updating policy.

⇒ **References for this Lesson:**

Sutton, R. S., Barto, A. G. (2018). Reinforcement learning: An introduction. MIT press. (Chapter 2)

TO-DO NEXT: Now, please go to the associated Jupyter Notebook of this Lesson to understand the coding implementation of the non-stationary k -Armed Bandit problem.

In the Next Lesson we will cover some fine-tuning techniques in the context of Armed-Bandit problems and will develop an application to a stock picking problem.